Sheshathri K

Software Development Engineer | Backend & Data Systems

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Professional Summary

Results-driven Software Development Engineer (SDE) with proven expertise across Backend Developer and Data Engineer roles. Experienced in designing high-throughput Java Spring Boot microservices, concurrent data processing pipelines, and Apache NiFi orchestrations on Kubernetes. Proven track record in optimizing REST API ingestion, eliminating distributed pipeline bottlenecks, and architecting scalable database solutions across MongoDB and SQL. Strong foundation in Data Structures, Algorithms, and scalable system design.

Technical Skills

- **Programming Languages:** Java (Core Java, Multithreading, Concurrency, Streams API, Collections), Python, JavaScript (ES6+), SQL, Bash/Shell Scripting
- **Data Engineering & Automation:** Apache NiFi, ETL/ELT Pipeline Design, n8n Workflow Automation, REST API Ingestion, Batch Data Processing, JSON/XML Parsing, Data Pipeline Monitoring
- **Backend Development:** Spring Boot, Microservices Architecture, RESTful API Design & Integration, Hibernate/JPA, OAuth 2.0, JWT & Token-Based Authentication, Django, Node.js, Express.js
- **Databases & Storage:** MongoDB, PostgreSQL, MySQL, SQL Query Optimization, Indexing Strategies, Schema Design, ACID Transactions, Connection Pooling
- **Cloud & DevOps:** Microsoft Azure, Kubernetes (Pod Orchestration, Workflows), Docker Containerization, CI/CD Pipelines, Linux/Unix Environments
- **Web Technologies:** HTML5, CSS3, JavaScript, React.js, Responsive Web Design, Modern CSS Grid/Flexbox, Cross-Browser Compatibility
- **Developer Tools & Frameworks:** Git, GitHub, Maven, Postman, FFmpeg (Audio/Video Processing), NLTK, TextBlob, Scikit-learn
- **Core Competencies:** System Design, Multi-threaded Processing, High-Throughput API Ingestion, Performance Optimization, Root Cause Debugging

Work Experience

Uniphore Software Systems **Chennai, India**
Data Engineering Intern *Mar 2024 – Jun 2024*

- Engineered a high-throughput Java Spring Boot microservice to ingest 1,000+ daily call records via NICE CXone REST APIs; integrated OAuth 2.0 token caching and cursor pagination, reducing redundant API requests by 30%.
- Architected a concurrent audio processing pipeline leveraging Java multi-threading (`ExecutorService`) and FFmpeg, accelerating batch conversion and filtering of 500+ daily audio files by 4x with zero data corruption.
- Constructed an automated batch ingestion service for downstream speech analytics APIs, implementing exponential backoff retry mechanisms and payload validation to elevate pipeline data integrity to 99.8%.
- Orchestrated enterprise Apache NiFi ETL pipelines deployed on Kubernetes, streaming real-time health telemetry across 5+ processing stages into MongoDB and establishing automated alerting for 40+ concurrent workflows.

Technical Projects

Chatbot Song Recommender System *Jun 2023 – Dec 2023*
Python, Django, Spotify API, MongoDB, SQL, NLTK, TextBlob

- Engineered an intelligent music recommendation engine combining collaborative and content-based filtering algorithms, delivering 92% recommendation relevance across 50+ active test users.
- Integrated NLP-driven mood and sentiment analysis using NLTK and TextBlob with Spotify Web API to dynamically generate personalized playlists, implementing an in-memory caching layer that reduced external API calls by 40%.
- Designed and optimized document schemas in MongoDB for track metadata and user mood history, achieving sub-second playlist generation latency and improving user session engagement by 35%.

Park Explorer – Smart Parking Reservation Platform *Jan 2023 – Apr 2023*
Python, SQL, Google Maps API, REST APIs

- Architected an end-to-end parking reservation platform managing real-time slot synchronization across 100+ parking locations, maintaining 99.2% system uptime for 500+ monthly active users.
- Implemented an optimized greedy routing algorithm integrated with Google Maps Distance Matrix API, reducing average commuter parking search duration by 35% and increasing space allocation accuracy by 28%.
- Designed ACID-compliant transactional booking workflows in SQL with row-level locking to resolve concurrency bottlenecks, eliminating double-booking conflicts by 98%.

Education

SRM University — **Chennai, India**
B.Tech in Computer Science and Engineering – CGPA: 8.5 / 10.0 — *Sep 2021 – May 2025*

Bharathidhasanar Matric Higher Secondary School — **Arakkonam, India**
Class XII (Higher Secondary) – Score: 88% (8.8 / 10.0) — *Jun 2019 – Mar 2021*

Bharathidhasanar Matric Secondary School — **Arakkonam, India**
Class X (Secondary School) – GPA: 7.9 / 10.0 — *Jun 2018 – Mar 2019*

Certifications & Technical Accreditations

- **Microsoft Azure:** Cloud Computing & Infrastructure (Course & Internship – Verzeo, 2023)
- **Amazon Web Services:** AWS Cloud Foundations
- **Professional Tracks:** Java Full-Stack Developer Specialization, Generative AI Bootcamp, Cisco Network Essentials

Languages

English (Professional Working Proficiency) | Telugu (Native) | Tamil (Conversational)